

Task 8

Task Title:

Build an AI Customer Support Ticket Automation System

Task Description:

In this task, you will build a Generative AI system that automatically analyzes customer support requests and processes them using backend functions.

The system will simulate a real customer support workflow where AI understands the customer's issue, classifies the request, generates a response, and performs automated actions.

The system should:

- Accept a customer support message
- Use an LLM to analyze and classify the issue
- Generate a professional response
- Automatically execute functions based on the issue
- Save the ticket information and processing result

Task Objectives:

- Load or call an LLM
- Generate structured outputs like category, priority, response, and status
- Create API-like functions for ticket operations
- Automate the complete support workflow
- Handle simple errors such as empty input or invalid classification

Task Scenario:

- User enters a customer complaint or support request
- AI identifies the issue category (Technical, Billing, Account, General)
- AI determines the priority (Low, Medium, High)
- AI generates a professional response
- The system automatically performs an action based on the category
- The system saves the ticket as a file or database record
- The system displays the final ticket information and AI response